

MERCON Logistics — Business Rules

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Overview

These business rules define the operational logic and constraints that govern the MERCON Logistics platform, ensuring consistency, data integrity, and a standardized experience across the Driver Operations App, Operator Mobile App, Operator Web Dashboard, and Admin Web Dashboard.

BR-001 · User Authentication & Access Control

Rule	Description
BR-001.1	Every user must authenticate using authorized credentials before accessing the system
BR-001.2	Admin & Operator authenticate using Email + Password
BR-001.3	Drivers authenticate using Employee ID or Mobile Number + PIN
BR-001.4	Users shall only have access to modules permitted by their assigned role
BR-001.5	Sessions must automatically expire after prolonged inactivity
BR-001.6	All user activities shall be recorded in the audit log
BR-001.7	JWT-based authentication with role-based access control enforced across all applications

BR-002 · Driver Assignment

Rule	Description
BR-002.1	A driver may only have one active trip at any given time
BR-002.2	Only drivers marked as Available may be assigned to a trip
BR-002.3	Drivers cannot manually assign or modify trips
BR-002.4	Assignments become immediately available in the Driver App with a push notification

BR-003 · Vehicle Assignment

Rule	Description
BR-003.1	A vehicle may only be assigned to one active trip at a time
BR-003.2	Vehicles under maintenance cannot be assigned
BR-003.3	Vehicles with expired documentation are unavailable for assignment
BR-003.4	Only vehicles with sufficient capacity may be assigned to the selected cargo

BR-004 · Trip Creation

Rule	Description
BR-004.1	Every trip must be associated with a registered customer
BR-004.2	Pickup and delivery locations are mandatory
BR-004.3	Cargo information must be completed before a trip can be created
BR-004.4	A driver and vehicle must be assigned before trip status becomes Assigned
BR-004.5	Each trip shall have a unique Trip ID
BR-004.6	Trips can be created from both the Operator Web Dashboard and Operator Mobile App
BR-004.7	Trip types supported: One-Time, Daily, Weekly, Monthly

BR-005 · Pricing & Quotations

Rule	Description
BR-005.1	The system shall automatically suggest pricing based on predefined route rates
BR-005.2	Operators may manually override the suggested price
BR-005.3	Customer-specific pricing takes precedence over default pricing
BR-005.4	Every pricing modification shall be recorded
BR-005.5	Operators may save quotations linked to transportation requests
BR-005.6	Invoicing is out of scope for MVP — only quotations are supported

BR-006 · Driver Workflow

Rule	Description
BR-006.1	If an active trip exists, the Driver App opens directly to the current trip
BR-006.2	Driver taps "Start Trip" to initiate the pre-departure checklist
BR-006.3	Drivers cannot proceed without completing mandatory cargo photo verification
BR-006.4	Trip status changes to In Transit after cargo photos are successfully uploaded
BR-006.5	The navigation screen automatically opens after the trip begins

BR-006.6	Drivers can trigger Emergency at any time via the FAB button during an active trip
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BR-007 · GPS Tracking

Rule	Description
BR-007.1	Driver mobile GPS is the primary source of live location tracking
BR-007.2	ICCES vehicle GPS functions as a secondary tracking source when available
BR-007.3	GPS tracking begins automatically when a trip starts and ends when it completes
BR-007.4	Location updates synchronize continuously with the Operator Dashboard

BR-008 · Emergency Handling

Rule	Description
BR-008.1	Drivers may report emergencies at any time during an active trip
BR-008.2	Emergency alerts shall immediately notify the assigned operator
BR-008.3	The driver's live GPS location shall accompany every emergency report
BR-008.4	Emergency events shall be stored within the trip history
BR-008.5	Trip status changes to Emergency / Halted

BR-009 · Arrival & Delivery Verification

Rule	Description
BR-009.1	The application shall detect arrival using GPS geofencing (default radius: 500 meters)
BR-009.2	Drivers receive an automatic "You Have Arrived" notification
BR-009.3	Fallback: Manual "I Have Arrived" button if geofence doesn't trigger within expected timeframe
BR-009.4	Delivery photographs are mandatory before completing a trip
BR-009.5	Trip status changes to Completed after successful delivery verification

BR-010 · Notifications

Rule	Description
BR-010.1	Auto-generated for: Trip Assigned, Started, Cargo Uploaded, Driver Arrived

BR-010.2	Auto-generated for: Emergency Reported, Trip Halted, Delivery Uploaded, Completed
BR-010.3	Auto-generated for: Vehicle Maintenance & Document Expiry Reminders
BR-010.4	Emergency notifications are delivered immediately with highest priority

BR-011 · Operator Operations

Rule	Description
BR-011.1	Operators may create, edit, and manage transportation requests from web and mobile
BR-011.2	Operators may assign or reassign drivers before a trip starts
BR-011.3	Operators receive real-time updates for every trip event
BR-011.4	Operators may generate operational reports at any time
BR-011.5	Operators may share trips via WhatsApp, PDF, or Link

BR-012 · Fleet Management

Rule	Description
BR-012.1	Every truck shall have a unique fleet record
BR-012.2	Maintenance schedules shall be recorded for every vehicle
BR-012.3	Vehicle documentation shall include expiry dates
BR-012.4	Vehicles with expired compliance documents are unavailable for assignment

BR-013 · Customer Management

Rule	Description
BR-013.1	Every customer shall have a unique customer profile
BR-013.2	Customer trip history is retained permanently unless archived
BR-013.3	Customer quotations are linked to transportation requests

BR-014 · Document Management

Rule	Description
BR-014.1	Trip photographs are permanently linked to their corresponding trip

BR-014.2	Driver and vehicle documents include expiry tracking
BR-014.3	Document renewal reminders are generated automatically

BR-015 · Reporting

Rule	Description
BR-015.1	Reports generated for Trips, Drivers, Fleet, Customers, Revenue, Pricing, and Performance
BR-015.2	Reports available daily, weekly, monthly, and for custom date ranges

BR-016 · Audit & Security

Rule	Description
BR-016.1	All critical system actions shall be logged
BR-016.2	Role-based permissions shall be enforced across all applications
BR-016.3	Sensitive operational data shall be securely stored and transmitted (HTTPS, password/PIN hashing)
BR-016.4	Deleted records remain recoverable per the organization's retention policy
BR-016.5	API rate limiting shall be enforced to protect against abuse

Business Rule Summary

Rule	Category
BR-001	Authentication & Access Control
BR-002	Driver Assignment
BR-003	Vehicle Assignment
BR-004	Trip Creation
BR-005	Pricing & Quotations
BR-006	Driver Workflow
BR-007	GPS Tracking
BR-008	Emergency Handling
BR-009	Arrival & Delivery Verification
BR-010	Notifications
BR-011	Operator Operations
BR-012	Fleet Management
BR-013	Customer Management
BR-014	Document Management

BR-015	Reporting
BR-016	Audit & Security